Diatica*

GLICLAZIDE TABLETS BP 80 mg

Composition:
Each uncoated tablet contains: Gliclazide BP 80mg.

Indication:
Used in the treatment of NIDDM

Product Description:
White to yellowish white uncoated tablets with an engraved central line on one side and other side plain.

Pharmacodynamic:

Mechanism of action
Gliclazide is a hypoglycaemic sulfonylurea antidiabetic active substance differing from other related compounds by an N-containing heterocyclic ring with an endocyclic bond.

Gliclazide reduces blood glucose levels by stimulating insulin secretion from the β-cells of the islets of Langerhans. Increase in postprandial insulin and C-peptide secretion persists after two years of treatment.
In addition to these metabolic properties, gliclazide has haemovascular properties.

Effects on insulin release:

In type 2 diabetics, gliclazide restores the first peak of insulin secretion in response to glucose and increases the second phase of insulin secretion. A significant increase in insulin response is seen in response to stimulation induced by a meal or glucose.

Haemovascular properties:
Gliclazide decreases microthrombosis by two mechanisms which may be involved in complications of diabetes:
- A partial inhibition of platelet aggregation and adhesion, with a decrease in the markers of platelet activation (beta thromboglobulin, thromboxane B_2),
- An action on the vascular endothelium fibrinolytic activity with an increase in tPA activity.

Pharmacokinetic:
Absorption
Plasma levels increase reaching maximal concentrations between 2 and 6 hours.
Gliclazide is well absorbed. Food intake does not affect the rate or degree of absorption.
Distribution
Plasma protein binding is approximately 95%. The volume of distribution is around 19 litres.
Biotransformation
Gliclazide is mainly metabolised in the liver and excreted in the urine; less than 1% of the dose is excreted unchanged in the urine. No active metabolites have been detected in plasma.
Elimination
The elimination half life of gliclazide is between 10 and 12 hours.
Special populations
Elderly
No clinically significant changes in pharmacokinetic parameters have been observed in elderly patients.

Side effects
The following undesirable effects have been reported.
The most frequent adverse reaction with gliclazide is hypoglycaemia. Possible symptoms of hypoglycaemia are: headaches, intense hunger, nausea, vomiting, lassitude,sleep disorders, agitation, aggression, poor concentration, reduced awareness and slowed reactions, depression, confusion, visual and speech disorders, aphasia, tremor, paresis, sensory disorders, dizziness, feeling of powerlessness, loss of self control, delirium, convulsions, shallow respiration, bradycardia, drowsiness and loss of consciousness, possible resulting in coma and lethal outcome.
In addition, signs of adrenergic counter-regulation may be observed: sweating, clammy skin, anxiety, tachycardia, hypertensions, palpitations, angina pectoris and cardiac arrhythmia.
Usually symptoms disappear after intake of carbohydrates (sugar). Experience with other sulfonylureas shows that hypoglycaemia can recur even when measures prove effective initially.
Gastrointestinal disturbances, including abdominal pain, nausea, vomiting, dyspepsia, diaarhoea, and constipation have been reported: if these should occur they can be avoided or minimised if gliclazide is taken with breakfast.
The following undesirable effects have been more rarely reported:
- Skin and subcutaneous tissue disorders: rash, pruritus, urticaria, angioedema, erythema, maculopapular rashes, bullous reactions (Such as Stevens-Johnson syndrome and toxic epidermal necrolysis and autoimmune bullous disorders), and exceptionally, drug rash with eosinophilia and systemic symptoms (DRESS).
- Blood and lymphatic system disorders: changes in haematology are rare. They may include anaemia, leucopenia, thrombocytopenia, granulocytopenia. These are in general reversible upon discontinuation of medication.
- Hepato-biliary disorders: raised hepatic enzyme (AST, ALT, alkaline phosphatases), hepatitis (isolated reports). Discontinue treatment if cholestatic jaundice appears. These symptoms usually disappear after discontinuation of treatment.
- Eye disorders:Transient visual disturbances may occur especially on initiation of treatment, due to changes in blood glucose levels.
- Class attribution effects:

As for other sulfonylureas, the following adverse events have been observed: cases of erythrocytopenia, agranulocytosis, haemolytic anaemia, pancytopenia, allergic vasculitis, hyponatremia, elevated liver enzyme levels and even impairment of liver function (e.g. with cholestasis and jaundice) and hepatitis which regressed after withdrawal of the sulfonylurea or led to life-threatening liver failure in isolated cases.

Overdose
An overdose of sulfonylureas may cause hypoglycaemia. Moderate symptoms of hypoglycaemia, without any loss of consciousness or neurological signs, must be corrected by carbohydrate intake, dose adjustment and/or change of diet. Strict monitoring should be continued until the doctor is sure that the patient is out of danger. Severe hypoglycaemic reactions, with coma, convulsions or other neurological disorders are possible and must be treated as a medical emergency, requiring immediate hospitalisation. If hypoglycaemic coma is diagnosed or suspected, the patient should be given a rapid I.V. injection of 50 mL of concentrated glucose solution (20 to 30 %). This should be followed by continuous infusion of a more dilute glucose solution (10 %) at a rate that will maintain blood glucose levels above 1 g/L. Patients should be monitored closely and, depending on the patient's condition after this time, the doctor will decide if further monitoring is necessary. Dialysis is of no benefit to patients due to the strong binding of gliclazide to proteins.

Contraindication
This medicine is contra-indicated in case of:
- Hypersensitivity to gliclazide or to any of the excipients, other sulfonylureas, sulfonamides,
- Type 1 diabetes
- Diabetic pre-coma and coma, diabetic keto-acidosis,
- Severe renal or hepatic insufficiency: in these cases the use of insulin is recommended,
- Treatment with miconazole
- Lactation

Warnings and Precautions:
Hypoglycaemia:
This treatment should be prescribed only if the patient is likely to have a regular food intake (including breakfast). It is important to have a regular carbohydrate intake due to the increased risk of hypoglycaemia if a meal is taken late, if an inadequate amount of food is consumed or if the food is low in carbohydrate. Hypoglycaemia is more likely to occur during low-calorie diets, following prolonged or strenuous exercise, alcohol intake or if a combination of hypoglycaemic agents is being used.
Hypoglycaemia may occur following administration of sulfonylureas . Some cases may be severe and prolonged. Hospitalisation may be necessary and glucose administration may need to be continued for several days. Careful selection of patients, of the dose used, and clear patient directions are necessary to reduce the risk of hypoglycaemic episodes.
Factors which increase the risk of hypoglycaemia:
• patient refuses or is unable to co-operate,
• malnutrition, irregular mealtimes, skipping meals, periods of fasting or dietary changes,
• imbalance between physical exercise and carbohydrate intake,
• renal insufficiency,
• severe hepatic insufficiency,
• overdose of Diatica 80 mg Tablets,
• certain endocrine disorders: thyroid disorders, hypopituitarism and adrenal insufficiency,
• concomitant administration of certain other medicines (see Interaction with Medicament).

Renal and hepatic insufficiency: the pharmacokinetics and/or pharmacodynamics of gliclazide may be altered in patients with hepatic insufficiency or severe renal failure. A hypoglycaemic episode occurring in these patients may be prolonged, so appropriate management should be initiated.

Patient information: the risks of hypoglycaemia, together with its symptoms, treatment, and conditions that predispose to its development, should be explained to the patient and to family members. The patient should be informed of the importance of following dietary advice, of taking regular exercise, and of regular monitoring of blood glucose levels.

Poor blood glucose control: blood glucose control in a patient receiving antidiabetic treatment may be affected by any of the following: St. John's Wort (*Hypericum perforatum*) preparations, fever, trauma, infection or surgical intervention. In some cases, it may be necessary to administer insulin. The hypoglycaemic efficacy of any oral antidiabetic agent, including gliclazide, is attenuated over time in many patients: this may be due to progression in the severity of the diabetes, or to a reduced response to treatment. This phenomenon
is known as secondary failure which is distinct from primary failure, when an active substance is ineffective as first-line treatment. Adequate dose adjustment and dietary compliance should be considered before classifying the patient as secondary failure.

Dysglycaemia:
Disturbances in blood glucose, including hypoglycaemia and hyperglycaemia have been reported, in diabetic patients receiving concomitant treatment with fluoroquinolones, especially in elderly patients. Indeed, careful monitoring of blood glucose is recommended in all patients receiving at the same time Diamicron 80 mg and a fluoroquinolone.

Laboratory tests: Measurement of glycated haemoglobin levels (or fasting venous plasma glucose) is recommended in assessing blood glucose control. Blood glucose self-monitoring may also be useful. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.
Treatment of patients with G6PD-deficiency with sulfonylurea agents can lead to haemolytic anaemia. Since gliclazide belongs to the class of sulfonylurea agents, caution should be used in patients with G6PD-deficiency and a nonsulfonylurea alternative should be considered.
Porphyric patients:
Cases of acute porphyria have been described with some other sulfonylurea drugs, in patients who have porphyria.

Drug Interaction
The following products are likely to increase the risk of hypoglycaemia

Contra-indicated combination
• **Miconazole** (systemic route, oromucosal gel): increases the hypoglycaemic effect with possible onset of hypoglycaemic symptoms, or even coma.

Combinations which are not recommended
• **Phenylbutazone** (systemic route): increases the hypoglycaemic effect of sulfonylureas (displaces their binding to plasma proteins and/or reduces their elimination).
It is preferable to use a different anti-inflammatory agent, or else to warn the patient and emphasise the importance of self-monitoring. Where necessary, adjust the dose during and after treatment with the anti-inflammatory agent.
• **Alcohol:** increases the hypoglycaemic reaction (by inhibiting compensatory reactions) that can lead to the onset of hypoglycaemic coma.
Avoid alcohol or medicines containing alcohol.

Combinations requiring precautions for use
Potentiation of the blood glucose lowering effect and thus, in some instances, hypoglycaemia may occur when one of the following drugs is taken:
other antidiabetic agents (insulins, acarbose, metformin, thiazolidinediones, dipeptidyl peptidase-4 inhibitors, GLP-1 receptor agonists), beta-blockers, fluconazole, angiotensin converting enzyme inhibitors (captopril, enalapril), H2-receptor antagonists, MAOIs, sulfonamides, clarithromycin and nonsteroidal anti-inflammatory agents.
The following products may cause an increase in blood glucose levels

Combination which is not recommended
• **Danazol**: diabetogenic effect of danazol.
If the use of this active substance cannot be avoided, warn the patient and emphasise the importance of urine and blood glucose monitoring. It may be necessary to adjust the dose of the antidiabetic agent during and after treatment with danazol.

Combinations requiring precautions during use
• **Chlorpromazine** (neuroleptic agent): high doses (>100 mg per day of chlorpromazine) increase blood glucose levels (reduced insulin release).
Warn the patient and emphasise the importance of blood glucose monitoring. It may be necessary to adjust the dose of the antidiabetic active substance during and after treatment with the neuroleptic agent.
• **Glucocorticoids** (systemic and local route: intra-articular, cutaneous and rectal preparations) and tetracosactrin: increase in blood glucose levels with possible ketosis (reduced tolerance to carbohydrates due to glucocorticoids).
Warn the patient and emphasise the importance of blood glucose monitoring, particularly at the start of treatment. It may be necessary to adjust the dose of the antidiabetic active substance during and after treatment with glucocorticoids.
• **Ritodrine, salbutamol, terbutaline**: (I.V.)
Increased blood glucose levels due to beta-2 agonist effects.
Emphasise the importance of monitoring blood glucose levels. If necessary, switch to insulin.
• **Saint John's Wort (*Hypericum perforatum*) preparations:**
Gliclazide exposure is decreased by Saint John's Wort-*Hypericum perforatum*. Emphasise the importance of blood glucose levels monitoring.
The following products may cause dysglycaemia
Combinations requiring precautions during use
• **Fluoroquinolones:** in case of a concomitant use of Diamicron 80 mg and a fluoroquinolone, the patient should be warned of the risk of dysglycaemia, and the importance of blood glucose monitoring should be emphasised.
Combination which must be taken into account
• **Anticoagulant therapy** (Warfarin ...):
Sulfonylureas may lead to potentiation of anticoagulation during concurrent treatment.
Adjustment of the anticoagulant may be necessary.

Pregnancy and Lactation:
Pregnancy:
There is no or limited amount of data from the use of gliclazide in pregnant women,
even though there are few data with other sulfonylureas.
As a precautionary measure, it is preferable to avoid the use of Gliclazide during pregnancy.
Control of diabetes should be obtained before the time of conception to reduce the risk of congenital abnormalities linked to uncontrolled diabetes.
Oral hypoglycaemic agents are not suitable, insulin is the drug of first choice for treatment of diabetes during pregnancy.
It is recommended that oral hypoglycaemic therapy is changed to insulin before a pregnancy is attempted, or as soon as pregnancy is discovered.
Breast-feeding:
It is unknown whether gliclazide or its metabolites are excreted in human milk. Given the risk of neonatal hypoglycaemia, the product is therefore contra-indicated in breast-feeding mothers. A risk to the newborns/infants cannot be excluded.
Fertility
No effect on fertility or reproductive performance was noted in male and female rats.

Dosages and administration
Gliclazide is recommended for the treatment of NIDDM in adults who have failed to respond to dietary restrictions. The daily dose generally required for adequate glycaemic control is 160mg but this may be increased to a maximum of 320mg/day. Dosages of over 160mg/day should be administered in 2 doses with morning and evening meals.

The dose may need to be titrated if a patient has been switched from a different brand of product containing Gliclazide.

Route of administration:
Oral

Storage:
Store below 30°C in a dry place. Protect from light.

Shelf life: 48 Months

Presentation:
Strip of 10 tablets

Manufactured by:
UNIQUE PHARMACEUTICAL LABORATORIES
(A Division of J.B. Chemicals & Pharmaceuticals Ltd.)
Plot No. 215-219, GIDC, Industrial Area, Panoli,
Bharuch, 394 116, India

Product Registration Holder & Imported by:
UNIMED SDN BHD
No. 53, Jalan Tembaga SD 5/2B, Bandar Sri Damansara,
52200 Kuala Lumpur, Malaysia.

Date of Revision: Dec. 2023

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360mm

130mm

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